

stat2011

Q7

```
set.seed(36748)
n = 10
B = 10000
popn = 1:6
n
```

```
## [1] 10
```

```
B
```

```
## [1] 10000
```

```
popn
```

```
## [1] 1 2 3 4 5 6
```

```
draws = sample(size = n*B, x = popn, replace = TRUE)

sample = matrix(draws, B, n)

dim(sample)
```

```
## [1] 10000    10
```

Q8

```
sim.sums = apply(sample, 1, sum)
mu = 3.5
mu
```

```
## [1] 3.5
```

```
sig.sq = 91/6 - 3.5^2  
sig.sq
```

```
## [1] 2.916667
```

```
sig = sqrt(sig.sq)  
sig
```

```
## [1] 1.707825
```

```
##(a)  
mean(sim.sums)
```

```
## [1] 35.031
```

```
n*mu
```

```
## [1] 35
```

```
##(b)  
var(sim.sums)
```

```
## [1] 29.22856
```

```
n*sig.sq
```

```
## [1] 29.16667
```

```
##(c)  
mean(sim.sums >= 45)
```

```
## [1] 0.0415
```

...